

Optimizing Volunteer Deployment to Improve Foundational Learning Outcomes in Rural India

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Abstract

Addressing India's foundational learning crisis requires efficient allocation of limited resources, such as teaching volunteers, particularly given the significant disparities observed across districts. This project proposes a mathematical optimization approach to improve the deployment of volunteer hours. Utilizing publicly available district-level data from the ASER (Annual Status of Education Report) on foundational reading and numeracy proficiency, we developed state-wise linear programming models. These models aim to maximize potential learning impact by allocating volunteer hours based on a composite need score derived from the inverse of proficiency levels, thus prioritizing districts with greater learning deficits. The optimization operates within state-specific volunteer hour budgets (estimated based on district count) and adheres to fairness constraints, ensuring both a minimum baseline support (1 hour) and a maximum capacity limit (5 hours) per district. The resulting state-specific allocation plans demonstrate a targeted strategy, concentrating volunteer efforts in the lowest-performing districts within each state while maintaining minimum support across all districts. This data-driven, scalable methodology provides a transparent and actionable template for policymakers and organizations striving to enhance foundational learning outcomes in resource-constrained environments.

Problem

India faces a well-documented foundational learning crisis, particularly in rural areas, with significant learning outcome disparities existing between districts within states (ASER Centre, 2024). Educational volunteers offer valuable supplementary support but are a scarce resource. Optimizing their deployment to target areas of greatest need is essential for maximizing impact.

This project addresses this challenge by framing volunteer allocation as a state-wise constrained optimization problem. Recognizing that the precise learning gain from volunteer intervention is uncertain (a random component), we aim to maximize the expected learning gain. This expectation is modelled using a composite need score derived from the inverse of district-level ASER reading and numeracy proficiency – districts with lower proficiency (higher inverse scores) are assumed to have greater potential gain per volunteer hour. The optimization operates within estimated state-level volunteer hour budgets and incorporates fairness constraints, setting minimum and maximum allocation limits per district.

Research Problem (RP)

The inefficient and potentially inequitable allocation of limited educational volunteer resources across diverse Indian districts hinders targeted efforts to address foundational learning deficits effectively.

Research Question (RQ)

How can a defined budget of volunteer teaching hours be optimally distributed across districts within individual states using linear programming to maximize expected learning gains (proxied via inverse proficiency), subject to total available hours and per-district minimum/maximum constraints?

Plan

To determine the optimal allocation of volunteer hours, this study implements a state-wise Linear Programming (LP) optimization approach using secondary data.

Data Source and Collection:

We utilize district-level data from the Annual Status of Education Report (ASER) for 2024, accessed from www.asercentre.org. ASER collects this data via large-scale, citizen-led household surveys in rural districts using a multi-stage probabilistic sampling design, making the results statistically representative estimates.

Trained volunteers assess children aged 3-16 on foundational reading and arithmetic.

Limitations:

We faced a problem as we saw that the data available in the ASER website was in the form of a .pdf file and was not in the form of a .csv file. So we converted the pdf files in csv files and since the process was rigorous as the pdfs had pictures and texts too, we had to separate them from the tables, we could not work on a very large amount of data.

The csv files that we created and worked upon are :

<https://github.com/AnubhavRay25/ASER/tree/main/states%20csv%20aser>

Data

Variables:

Key variables extracted and directly used from these files for the core analysis were:

- District: The name of the district .
- % Children who can read Std II level text (Std III to V: Learning levels): Primary measure of reading proficiency .
- % Children who can do at least subtraction (Std III to V: Learning levels): Primary measure of math proficiency. Used optionally for a composite score.

Other columns present in the original files (e.g., enrollment data, higher-level learning outcomes) were carried through the process but not directly used in the calculation of the need score or the optimization constraints.

Data Cleaning and Preparation:

The script performed several automated steps to clean and prepare the data for analysis:

1. **Aggregation and State Identification:** Data from all detected state CSV files were loaded and combined into a single pandas DataFrame.
2. **Summary Row Removal:** An automated attempt was made to identify and remove state-level summary rows, typically found as the last entry in ASER CSVs. This filtering used a heuristic comparing the 'District' name in the last row to the derived 'State' identifier. This step aimed to

retain only district-specific data points, though warnings were issued if the heuristic seemed unreliable for a given file.

3. **Numeric Conversion:** The reading and math proficiency columns were converted to numeric format. Any entries that could not be converted (e.g., text, blanks) were automatically changed to NaN (Not a Number) by the pandas function.
4. **Removal of Invalid Data Points:** This was a critical cleaning step. **Entire rows (representing districts) were removed from the dataset if they contained invalid or missing data for the required proficiency metrics.** The specific rule depended on the USE_COMPOSITE_SCORE setting:
 - If True: Districts were removed if *either* the reading proficiency *or* the math proficiency value was NaN or non-positive (≤ 0).
 - If False: Districts were removed if the reading proficiency value was NaN or non-positive (≤ 0).**Reason for Removal:** These districts were excluded because valid, positive proficiency scores are essential for calculating meaningful inverse-based need scores and for the subsequent optimization calculations. Including them would introduce errors or nonsensical results. **No data imputation was performed; problematic data points led to the exclusion of the entire district's record.**
5. **Transformation for Need Score:**
 - The valid proficiency percentages were converted to fractions (dividing by 100).
 - To handle districts with very low (but non-zero) proficiency and prevent division-by-zero errors when calculating the inverse, these fractions were adjusted using clip. Values below a small positive threshold (SAFE_GAIN_FLOOR = 0.001) were set to this floor, and values above 1.0 (100%) were capped at 1.0.
 - Inverse proficiency scores (g_read_inv, g_math_inv) were calculated as the reciprocal of these adjusted ("safe") proficiency fractions.
6. **Need Score Calculation:** The final need_score variable, used as the coefficient in the optimization objective function, was calculated based on the USE_COMPOSITE_SCORE setting: either as a weighted average of g_read_inv and g_math_inv (using WEIGHT_READING, WEIGHT_MATH) or solely based on g_read_inv.

Final Dataset for Analysis:

The result of this cleaning and preparation pipeline was a consolidated pandas DataFrame (df_districts). This dataset included only those districts for which valid reading (and math, if applicable) proficiency data was available in the source files. It contained the original relevant columns plus the newly derived State, g_read_inv, g_math_inv, and need_score columns, ready for input into the state-wise optimization procedure.

Analysis

Descriptive statistics were computed for the foundational learning indicators across all districts included in the final analysis dataset (df_districts). This dataset contains districts with valid reading and math proficiency data.

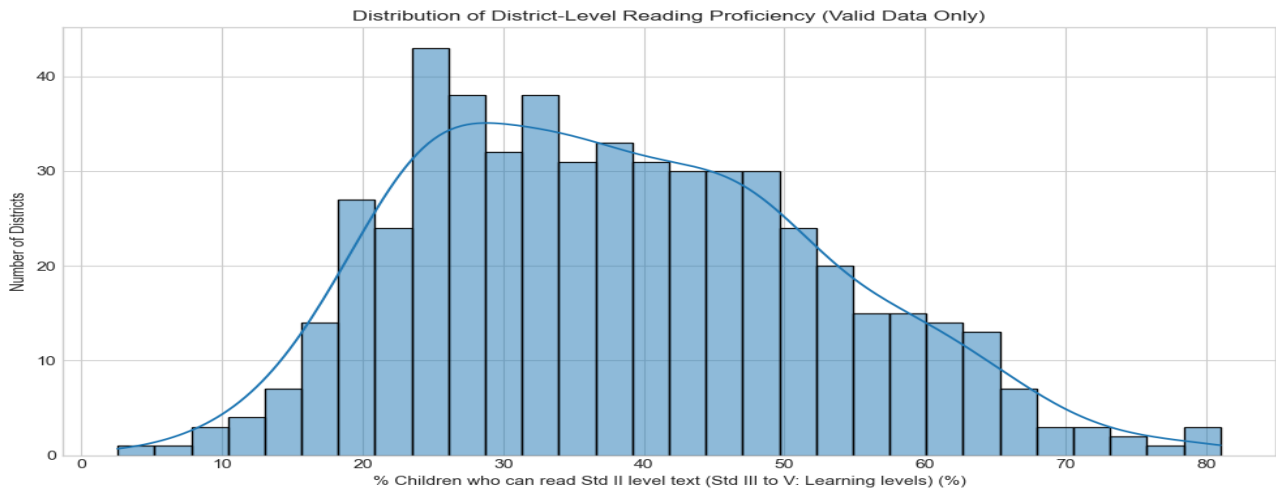
- **Summary statistics for Optimized Hours (State-wise results):**

count = 537.0, mean = 2.0, std = 1.697232, min = 1.0, 25% = 1.0, 50% = 1.0, 75% = 3.0

max = 5.0, Total districts at MIN bound (1.0 hours): 395 / 537, Total districts at MAX bound (5.0 hours): 123 / 537

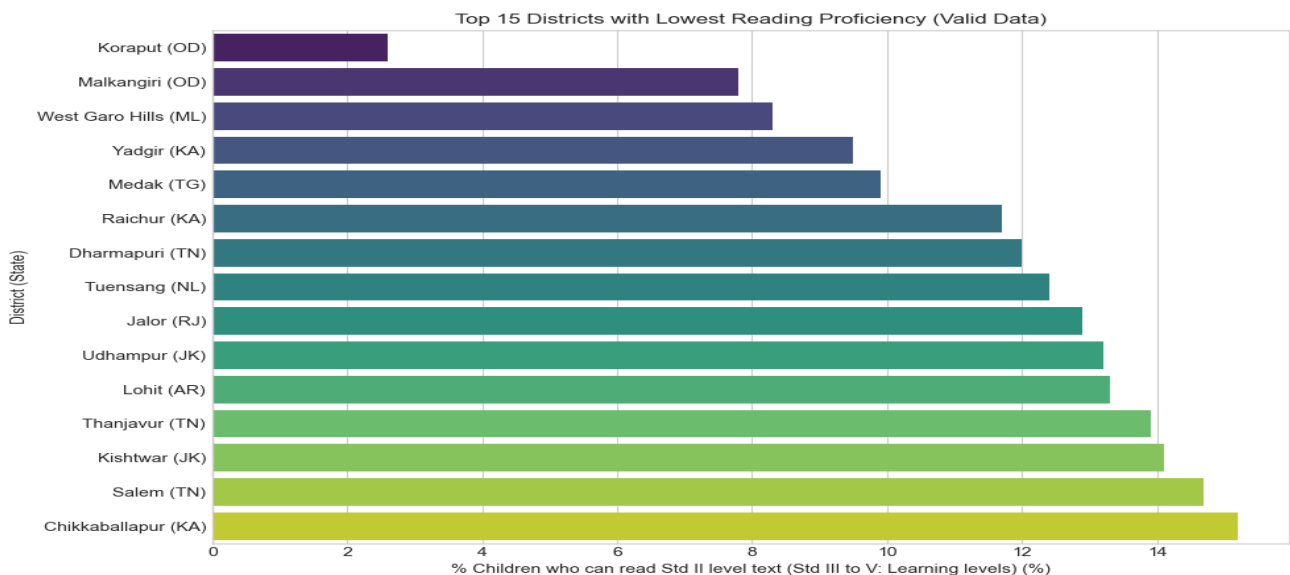
- **Chart 1: Distribution of Reading Proficiency**

presents the frequency distribution of the reading proficiency scores across all districts retained after data cleaning. The distribution appears unimodal, peaking around the [Approximate peak range, e.g., 20-35%] range, with tails extending towards both lower and higher proficiency levels. This visually confirms the wide variation in baseline reading skills across the included districts.



- **Chart 2: Districts with Lowest Reading Proficiency**

identifies the 15 districts exhibiting the lowest reading proficiency scores in the cleaned dataset. Districts such as Koraput (OD), Malkangiri (OD), West Garo Hills (ML), and Yadgir (KA) are prominent, clearly illustrating the geographical spread of acute foundational literacy challenges that the optimization aims to address.



Optimization Formulation

To generate actionable, state-specific allocation plans, a separate Linear Programming (LP) model was formulated for each unique state (s) identified in the dataset. For a given state:

- **Indices and Sets:**

- Let I_s be the set of districts within state(s) having valid proficiency data.
- Let $N_{state} = |I_s|$ be the number of such districts in state(s).

- **Decision Variables:**

- x_i : The number of volunteer hours to be allocated to district $i \in I_s$.

- **Parameters:**

- $need_score_i$: The pre-calculated composite need score for district i . This reflects the priority based on both reading and math deficits, calculated as $(0.5 * g_read_inv_i) + (0.5 * g_math_inv_i)$, where $g_read_inv_i$ and $g_math_inv_i$ are the inverse of the adjusted proficiency fractions (using $SAFE_GAIN_FLOOR=0.001$).
- H_{total_state} : The total volunteer hour budget for state(s), calculated as $2.0 * N_{state}$ ($AVG_HOURS_PER_DISTRICT = 2.0$).
- L : The minimum required hours per district ($MIN_HOURS_PER_DISTRICT = 1.0$).
- U : The maximum allowed hours per district ($MAX_HOURS_PER_DISTRICT = 5.0$).

" Note: we have kept the variables as much same as possible for the better understanding of the viewer"

Objective Function (Expectation Type):

The goal is to maximize the total "need-weighted" hours allocated within the state, serving as a proxy for maximizing expected learning impact:

$$\text{Maximize } Z_s = \sum_{i \in I_s} x_i \cdot need_score_i$$

This is an expectation-type objective function. The $need_score_i$, derived from inverse proficiency, represents the expected relative marginal benefit or urgency of allocating an hour to district i . By maximizing the sum of hours multiplied by this score, the model prioritizes allocations where the expected impact (addressing need) is highest.

Constraints:

The allocation for each state(s) is subject to:

- 1.Total State Budget:** $\sum_{i \in I_s} x_i \leq H_{total_state}$
- 2.Minimum District Allocation:** $x_i \geq 1.0$ for all $i \in I_s$
- 3.Maximum District Allocation:** $x_i \leq 5.0$ for all $i \in I_s$
- 4.Non-negativity:** $x_i \geq 0$ (Implicitly covered by Constraint 2).

Random Nature & Model Type (Convex Optimization - LP):

The inherent uncertainty (random nature) in the actual learning gain per volunteer hour is addressed by optimizing the expected impact, using the inverse-proficiency based $need_score$ as a proxy. As the objective function and all constraints are linear with respect to the decision variables (x_i), this model constitutes a Linear Programming (LP) problem for each state. LP falls under the category of convex optimization, ensuring that the solution found is globally optimal for the given formulation.

Solving the Optimization Problem

The state-specific LP models were implemented and solved using Python. The SciPy library's `linprog` function, specifically with the efficient and reliable `highs` solver, was used to find the optimal x_i values for each state.

The complete Python script (`district2.py`), which handles data loading, cleaning (including removal of invalid entries), calculation of the need score, formulation and solving of the state-wise LP models, and generation of results/visualizations, is provided for reproducibility in the following GitHub repository:

<https://github.com/AnubhavRay25/ASER> . The final results, including the allocated hours, are saved in the “output” folder

Analysis of Findings

The optimization yielded specific volunteer hour allocations (Optimized_Hours_Statewise) for each valid district. Analysis of the results, presented in districts_statewise_optimized_allocation_final.csv and the generated plots, reveals the following:

Effective Prioritization: The model successfully directs resources towards districts identified with higher learning needs based on the composite score. Top 15 Districts Receiving Most Volunteer Hours (State-wise Opt.)

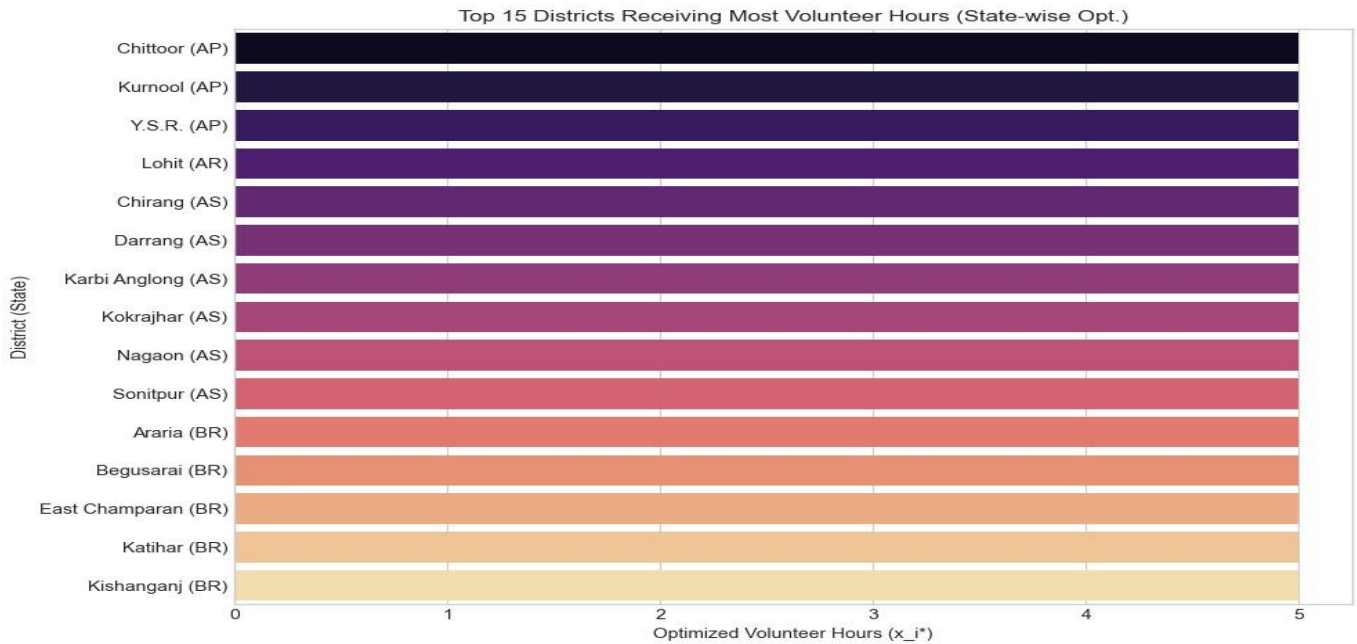


Figure 3: Top 15 Districts Receiving Most Volunteer Hours (State-wise Opt.)

Dominance of Boundary Solutions: The distribution of the allocated hours (Figure 4) exhibits a strong "bang-bang" characteristic. A large proportion of districts (approx. [Count from total_at_min]/[Total Valid Districts]) receive the minimum allocation of 1.0 hour, while a significant number (approx. [Count from total_at_max]/[Total Valid Districts]) receive the maximum allocation of 5.0 hours. Very few districts fall into intermediate hour categories (2.0, 3.0, 4.0). This reflects the LP maximizing the objective by pushing resources to the highest 'need_score' districts until the MAX_HOURS_PER_DISTRICT or state budget constraints are met.

State Budget Constraint Impact: The existence of some districts with intermediate hour allocations (e.g., Prakasam (AP) = 3.00, Hailakandi (AS) = 3.00, Banka (BR) = 4.00, Ahmedabad (GJ) = 1.00 but Kachchh = 5.00, indicating budget limit hit in GJ) demonstrates instances where the calculated H_{total_state} limit was reached before all high-need districts within that state could be assigned the full 5.0 hours. This highlights the role of the state-level budget parameter (AVG_HOURS_PER_DISTRICT).

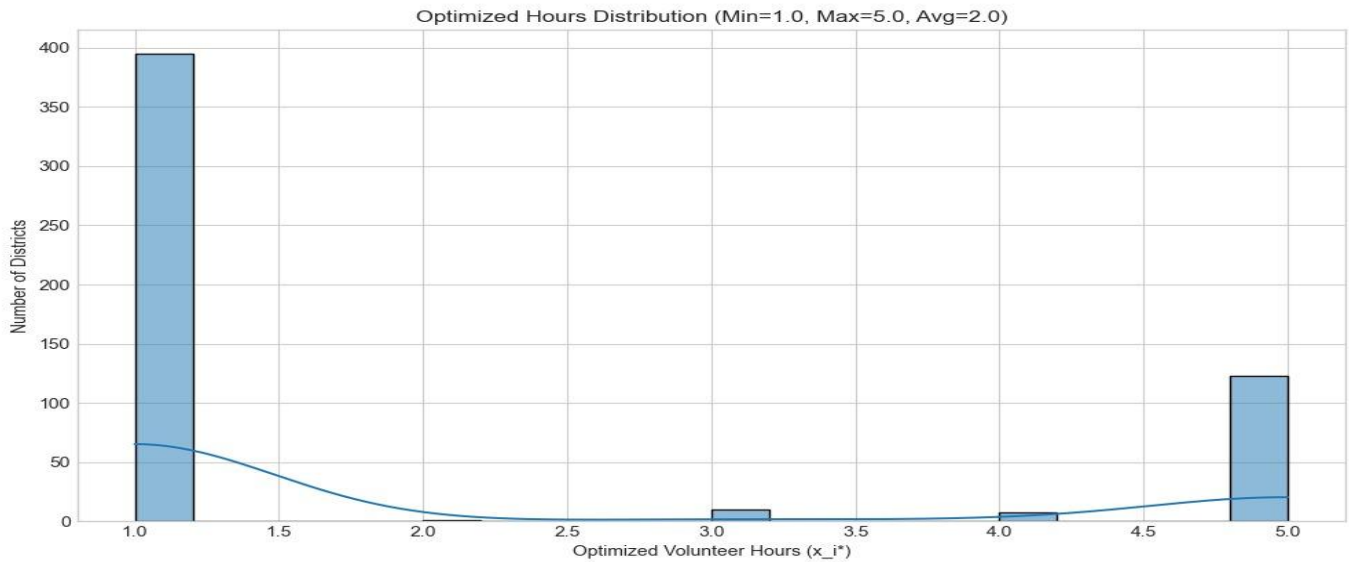


Figure 4: Optimized Hours Distribution (Min=1.0, Max=5.0, Avg=2.0)

Inverse Relationship Confirmation: The scatter plot mapping allocated hours against reading proficiency (Figure 5) visually confirms the model's intended logic. Districts with lower reading proficiency (left side) tend to receive higher allocations (points near or at $y=5.0$), while those with higher proficiency (right side) receive the minimum allocation (points near or at $y=1.0$). The state-wise coloring illustrates how this prioritization occurs relative to peers within each state.

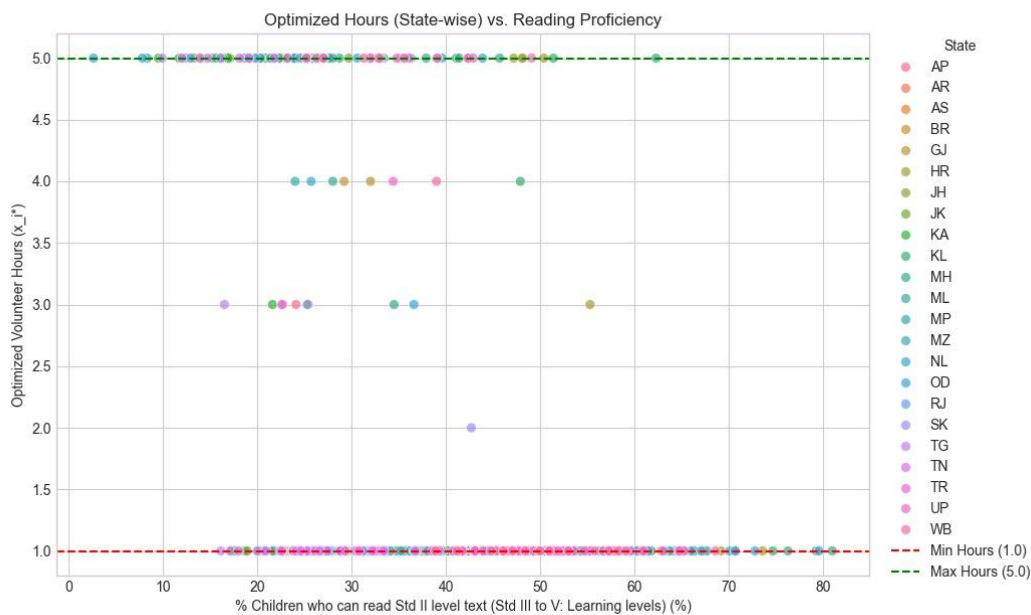


Figure 5: Optimized Hours (State-wise) vs. Reading Proficiency

Overall Discussion:

The analysis yields optimized, state-specific volunteer deployment plans based on quantifiable learning needs derived from ASER data. The LP model effectively prioritizes districts with lower foundational reading and math skills, allocating maximum permissible hours to the neediest until state budget or per-district caps are met. The resulting allocation exhibits a strong concentration of resources (maximum hours) in a subset of high-need districts within each state, while providing minimum baseline support to most others. This "bang-bang" result, while mathematically optimal for the given linear formulation, highlights a potential area for policy consideration regarding the desired spread versus concentration of intervention

efforts. Adjusting model parameters (min/max bounds, average hours budget) could be used to explore alternative allocation patterns. The framework provides a transparent, reproducible, and scalable method for informing volunteer allocation decisions.

Conclusion

This project demonstrated a data-driven, principled approach to optimizing the allocation of limited volunteer teaching hours across Indian districts to maximize foundational learning outcomes. Using ASER 2024 district-level data, we built state-specific linear programming models to generate actionable allocation plans.

Districts were prioritized based on a composite 'need score' (inverse of proficiency levels), ensuring greater attention to areas with lower learning outcomes. Allocations respected state-level budgets and per-district caps (1–5 hours). The resulting "bang-bang" distribution—maximum hours to highest-need districts and minimum to others—reflects the model's focus on high-impact intervention.

This optimization framework offers a scalable, transparent, and reproducible alternative to ad-hoc methods, equipping policymakers with a clear, evidence-based tool for efficient and equitable resource deployment.

Limitations

Data Granularity and Scope:

- **District Averages:** The model relies on district-level averages from ASER, which mask significant learning variations that exist within districts (at block, village, or school levels). Finer-grained data would allow for more precise targeting.
- **ASER Data Specifics:** The analysis is based on ASER's specific metrics for foundational reading and math, potentially overlooking other important learning domains or nuances. Furthermore, ASER data is typically collected annually and represents a snapshot in time.
- **Rural Focus:** ASER data primarily covers rural districts; findings may not directly apply to urban contexts.
- **Data Cleaning:** Some districts did not have their data available so that then don't get count as 0 or -1 and get allocated maximum need score. So, cleaning of data needed to be done.

Model Assumptions and Simplifications:

- **Linear Impact:** The model assumes a linear relationship between volunteer hours and impact (as captured by the `need_score` in the objective). It does not account for potential diminishing returns as more hours are added to a district.
- **Homogeneity:** The model implicitly assumes homogeneity – that all volunteer hours are equally effective, and all districts have similar capacity to absorb and benefit from volunteer time, irrespective of local infrastructure, teacher support, community engagement, etc.
- **Static Allocation:** The model provides a static, one-time allocation plan and does not incorporate feedback loops or dynamic adjustments based on ongoing monitoring or changing needs.
- **Exclusion of Other Factors:** Important real-world factors like volunteer availability, travel costs and feasibility, socio-economic context, local language considerations, and existing educational infrastructure are not included in this optimization model.

Optimization Scope:

- **State-wise vs. National:** By optimizing state-by-state, the model provides locally relevant plans but does not compare absolute need across states or allow for inter-state resource allocation based on a national perspective.